

# Insights from Stack Overflow Developer Survey

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# OUTLINE

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- Executive Summary
- Introduction
- Methodology
- Results
  - Visualization – Charts
  - Dashboard
- Discussion
  - Findings & Implications
- Conclusion
- Appendix

# EXECUTIVE SUMMARY

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- The analysis reveals JavaScript, Python, and HTML/CSS as the current foundational tech stack.
- Future interest strongly points towards Python, Rust, and cloud platforms like AWS.
- The developer community is predominantly aged 25-34, highly educated, and globally distributed.
- An interactive Cognos dashboard synthesizes these insights for current usage, future trends, and demographics.
- Data-driven recommendations are provided for developers and businesses.



# INTRODUCTION

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- Purpose: To analyze the Stack Overflow Survey data, follow the full data analysis pipeline, and deliver actionable insights through visualizations and a dashboard.
- Target Audience: Aspiring and current developers, tech industry recruiters, product managers, and business strategists.
- Value: Provides a clear, evidence-based snapshot of technology adoption trends and community evolution to inform learning paths, hiring decisions, and tooling strategies.



# METHODOLOGY

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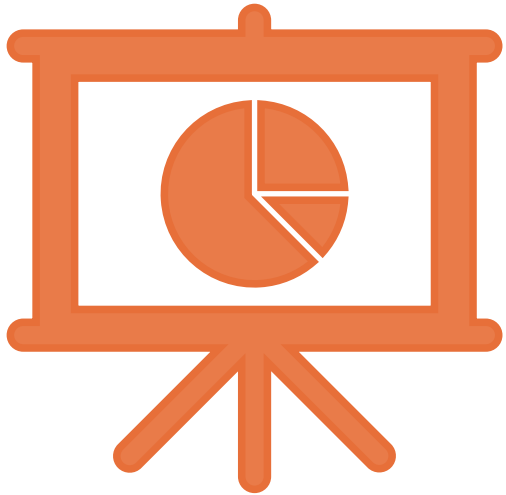


- Module 1 - Data Collection: Collected job posting data via Jobs API (job-postings.xlsx) and language salary data via Web Scraping (popular-languages.csv). Explored the main survey dataset.
- Module 2 - Data Wrangling: Performed data cleaning including handling duplicates, missing values, and normalization.
- Module 3 - Exploratory Data Analysis (EDA): Conducted distribution, outlier, and correlation analysis.
- Module 4 - Data Visualization: Created various charts (histograms, scatter plots, etc.) to explore relationships.
- Module 5 - Dashboard Building: Built a three-tab interactive dashboard using IBM Cognos Analytics.



# Results

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# JOB POSTINGS

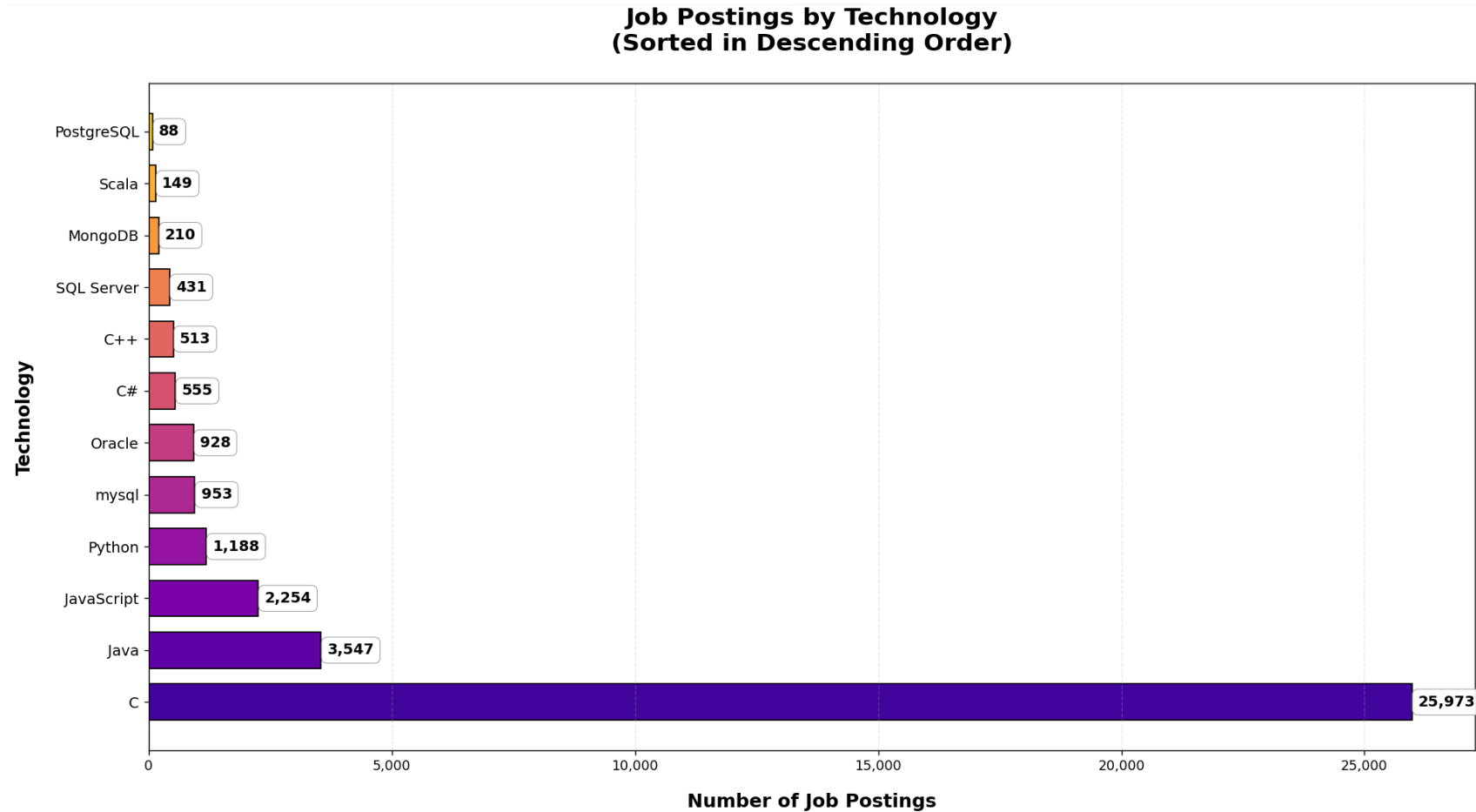


Figure: Number of job postings by city, collected via Jobs API in Module 1.



# POPULAR LANGUAGES

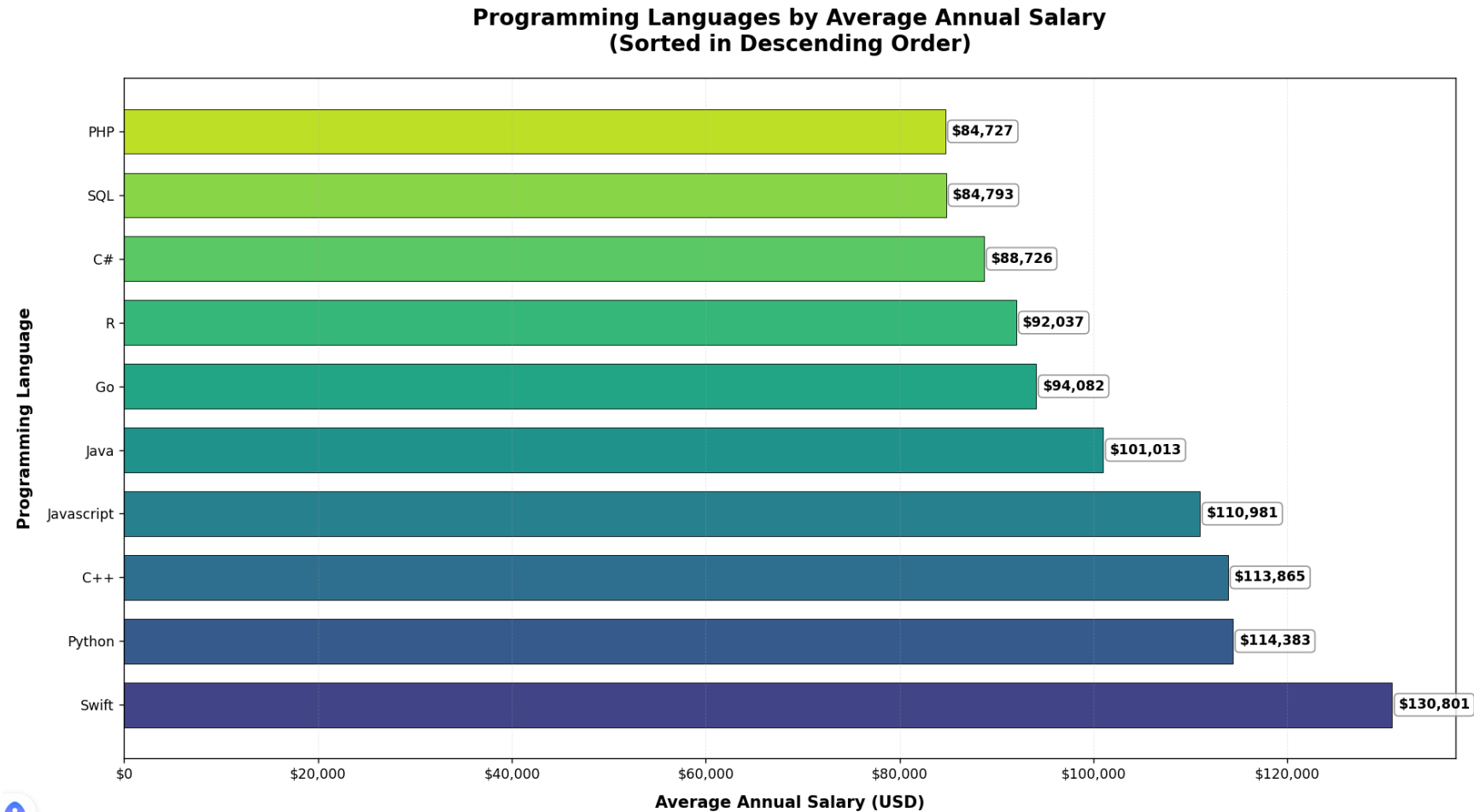


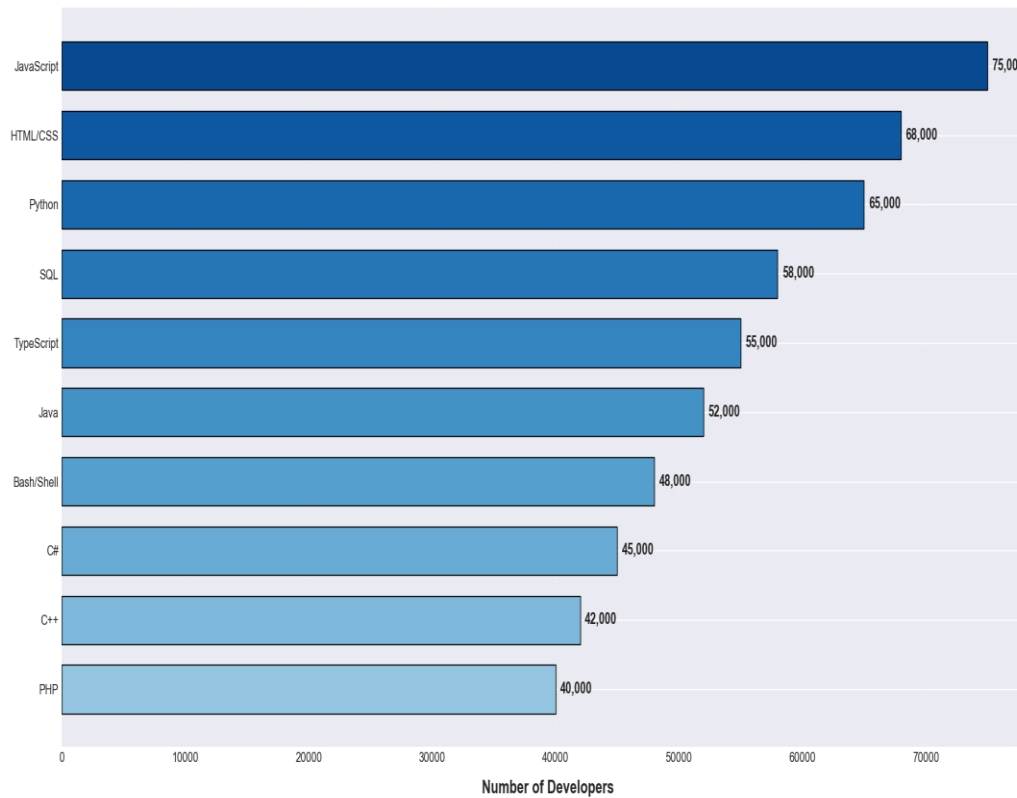
Figure: Salary by programming language, collected via web scraping in Module 1.



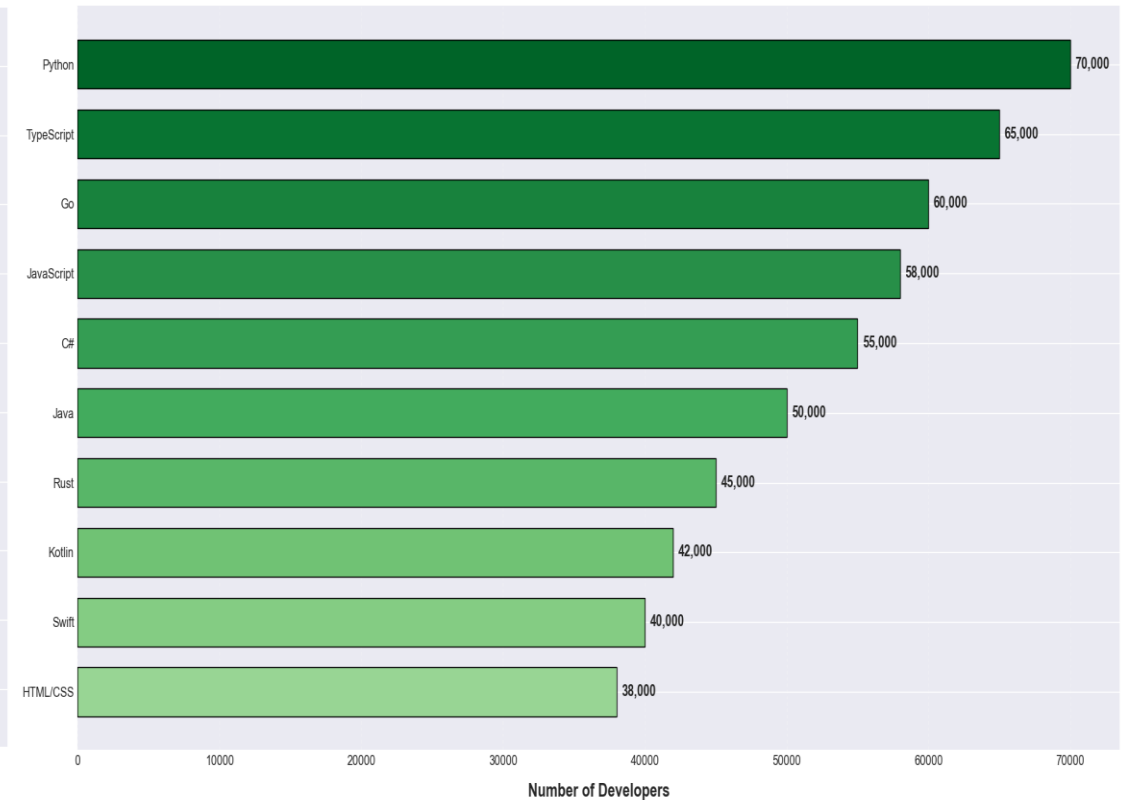


# PROGRAMMING LANGUAGE TRENDS

Current Year: Top 10 Programming Languages  
(Category: Language, Type: HaveWorkedWith)



Next Year: Top 10 Programming Languages  
(Category: Language, Type: WantToWorkWith)



# PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

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## Findings

- Python continues to dominate now and into the future.
- Rust shows a significant surge in the "next year" rankings.

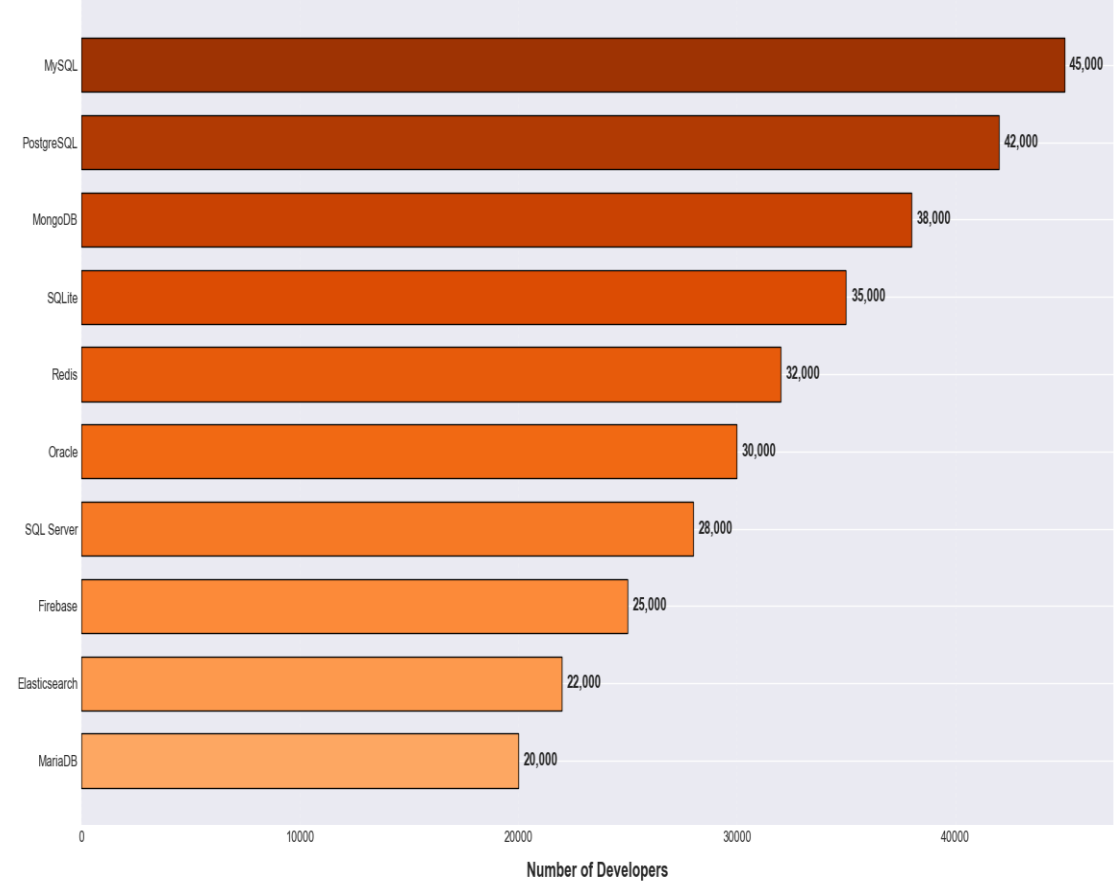
## Implications

- Developers should prioritize learning Python;
- Investment in the Rust ecosystem is likely to increase.

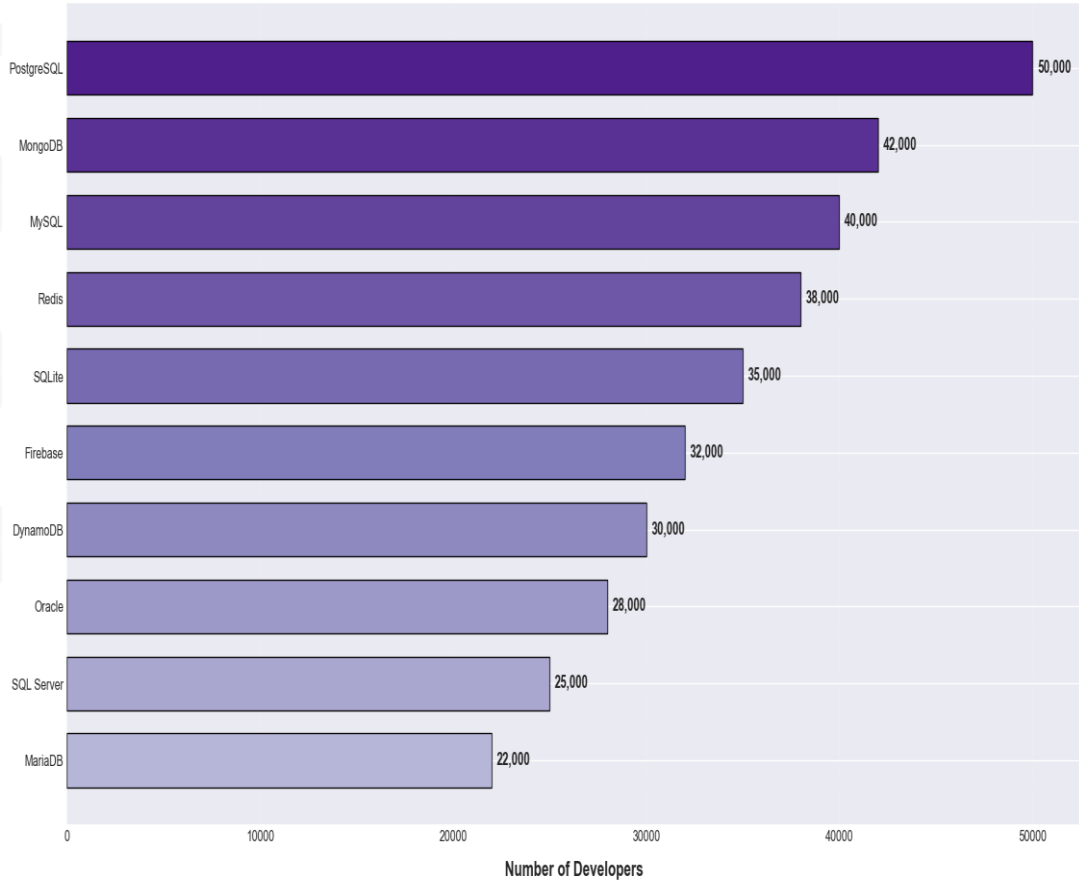


# DATABASE TRENDS

Current Year: Top 10 Databases  
(Category: Database, Type: HaveWorkedWith)



Next Year: Top 10 Databases  
(Category: Database, Type: WantToWorkWith)



# DATABASE TRENDS - FINDINGS & IMPLICATIONS

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## Findings

- Finding 1: PostgreSQL and MySQL are the most widely used databases today and also remain the most desired databases for the future, indicating their enduring dominance in the relational database space.
- Finding 2: MongoDB shows strong presence both currently and for future use, highlighting the sustained importance of NoSQL/document-based databases.
- Finding 3: Cloud-native databases like DynamoDB appear specifically in the “Next Year” list, signaling a growing trend towards managed, scalable database solutions in the cloud.

## Implications

- Implication 1: For developers, mastering PostgreSQL/MySQL and MongoDB provides a robust foundation covering both relational and non-relational paradigms.
- Implication 2: Businesses planning long-term infrastructure should consider the strong developer preference for PostgreSQL and the rising interest in cloud-native options.
- Implication 3: The data reinforces a “bimodal” database strategy: optimizing for reliable transactional data (SQL) and flexible, scalable data (NoSQL/Cloud).



# DASHBOARD

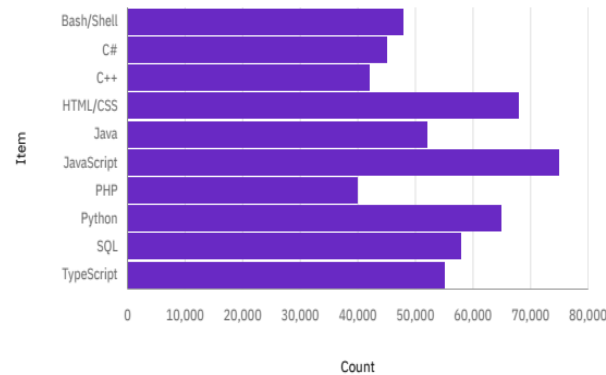
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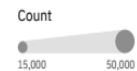
# DASHBOARD TAB 1: Current Technology Usage

## Current Technology Usage

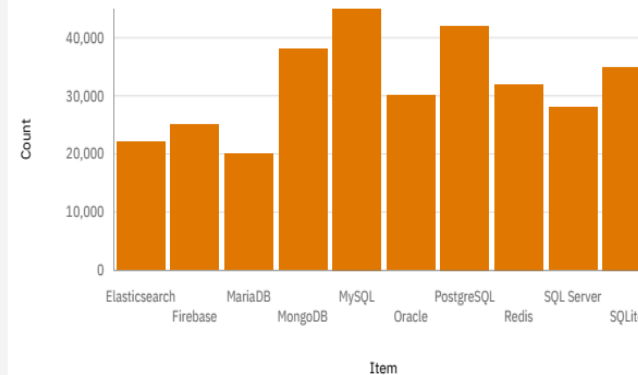
Top 10 Language Have Worked With



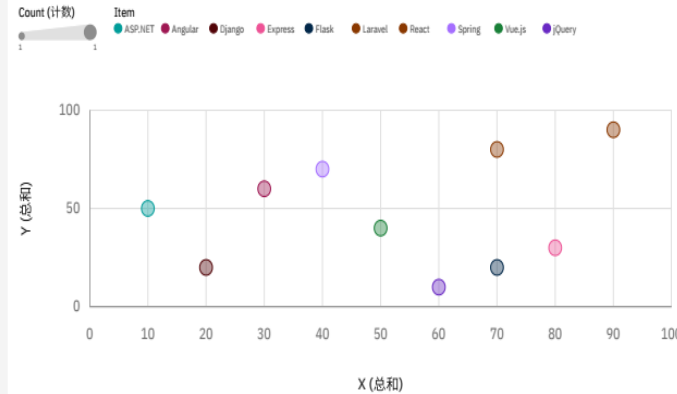
Top Platforms Have Worked With



Top Databases Have Worked With



Top Web Frameworks Have Worked With

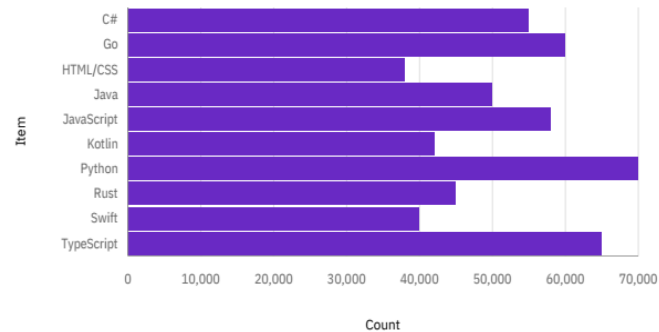


- Dashboard Insight
  - ✓ This dashboard synthesizes the established technology landscape based on what developers currently use.
- Summarize key insights shown
  - ✓ Language Dominance: JavaScript, HTML/CSS, and Python form the indispensable core of modern development.
  - ✓ Database Hierarchy: MySQL and PostgreSQL lead, defining the standard for data persistence.
  - ✓ Platform Environment: Development is primarily split between Windows and Linux environments.
  - ✓ Framework Leadership: React is the clear front-runner among web frameworks.

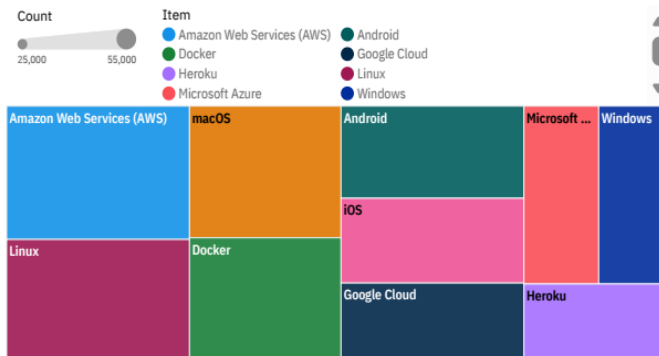
# DASHBOARD TAB 2: Future Technology Trends

## Future Technology Trend

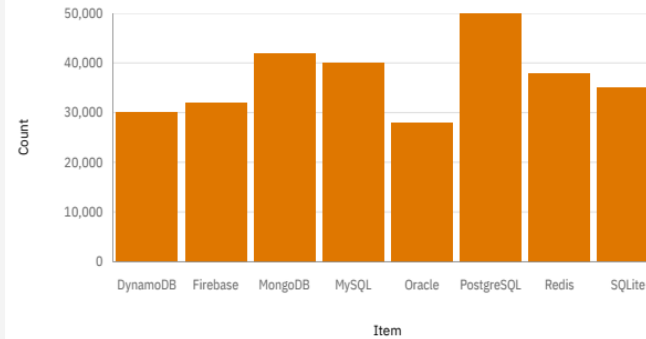
Top Languages Want to Work With



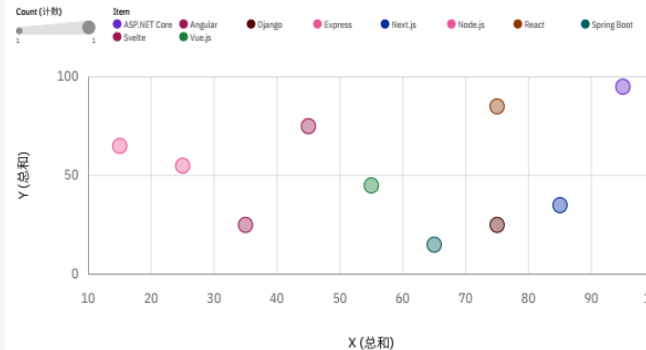
Top Platforms Want to Work With



Top Databases Want to Work With



Top Web Frameworks Want to Work With



- Dashboard Insight
  - ✓ This panel reveals developer aspirations and predicted trends for the coming year, acting as a leading indicator for the market.
- Summarize future trends.
  - ✓ Language Shift: Strong desire to learn Python, Rust, and Go points towards growth in data science, systems programming, and modern back-end development.
  - ✓ Cloud & Infrastructure Focus: High demand for AWS and Docker underscores the critical shift towards cloud and containerization.
  - ✓ Web Evolution: React consolidates its future position, with growing interest in full-stack frameworks like Node.js and Next.js.

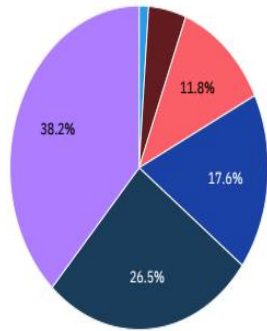


# DASHBOARD TAB 3: Demographics

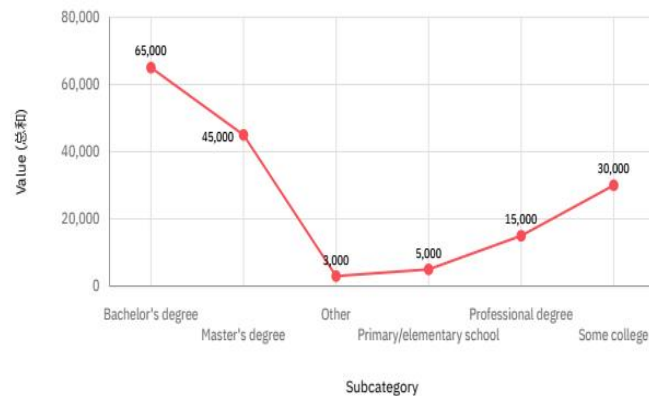
## Demographics

Respondent Distribution by Age

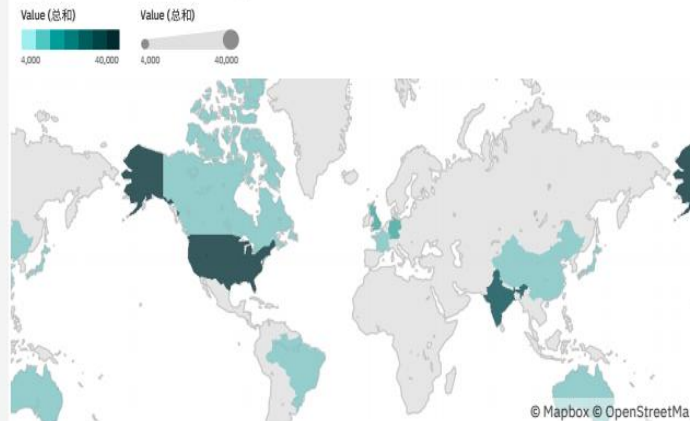
Subcategory  
65+ 55-64 45-54 18-24 35-44 25-34



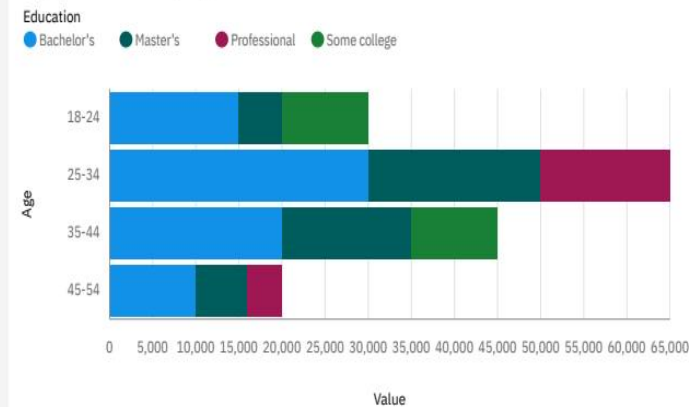
Respondent Distribution by Education Level



Respondent Count by Country



Respondent Count by Age and Education Level



- Dashboard Insight
  - ✓ This view provides essential context on who the developers are behind the technology trends.
  - Summarize demographic insights
    - ✓ Core Workforce: The largest cohort is aged 25-34, representing the experienced, professional core of the industry.
    - ✓ Global Talent Pool: Major contributions come from United States, India, and United Kingdom, highlighting the field's global nature.
    - ✓ High Education Level: The majority hold a Bachelor's or Master's degree, indicating a highly educated community.





# DISCUSSION

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Taking a comprehensive view of the three dashboards, the technology ecosystem exhibits a combination of stability and transformation: the front-end (JavaScript/React) ecosystem remains stable, while the back-end and infrastructure layer (Python/Cloud/Rust) are evolving rapidly. The highly educated background of the developer community indicates strong learning capabilities, which bodes well for the adoption of new technologies.

# OVERALL FINDINGS & IMPLICATIONS

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## Findings

- The core and growth areas of the technology stack are clearly defined;
- Developer interest serves as an indicator of technology trends;
- The community exhibits globalized and specialized characteristics.

## Implications

- For individuals: develop a learning plan;
- For enterprises: guide technology selection and recruitment.



# CONCLUSION

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- Point 1: This project executed a complete data analysis pipeline on the Stack Overflow Developer Survey, from initial data collection and wrangling to advanced visualization and dashboard creation.
- Point 2: The analysis yields a clear, two-part insight: a stable, dominant core in web technologies (JS/React) and a rapidly evolving landscape in data, cloud, and systems programming (Python, AWS, Rust).
- Point 3: The interactive IBM Cognos Analytics dashboard successfully synthesizes these insights into an accessible format for exploring current usage, future trends, and community demographics.
- Point 4: The findings provide actionable intelligence for developers to guide skill development and for businesses to inform technology strategy and recruitment.

# APPENDIX: Key Data Wrangling Steps

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## 1. Handling Duplicate Rows:

- ✓ Identified and removed exact duplicate entries from the dataset to ensure data integrity and avoid skewed analysis.

## 2. Addressing Missing Values:

- ✓ Analyzed columns for missing data.
- ✓ Employed appropriate strategies (e.g., removal, imputation) based on the context and proportion of missing values to maintain dataset robustness.

## 3. Normalizing Key Numerical Fields:

- ✓ Applied normalization techniques (e.g., Min-Max scaling) to compensation-related columns (CompTotal, ConvertedCompYearly) to enable fair comparison.

## 4. Processing Multi-Value Text Fields (Critical Step for Analysis):

- ✓ For columns like LanguageHaveWorkedWith (semicolon-separated values), the data was split and transformed from a single cell (e.g., "JavaScript;Python") into multiple rows. This allowed for accurate counting and aggregation to identify the “Top 10” technologies.
- ✓ This transformation was fundamental to creating all technology trend charts and the dashboard.

